

A Packing of 17 Four-Spaces in $\mathbb{R}^{11}$

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Abstract

We give 17 4-dimensional subspaces of $\mathbb{R}^{11}$ whose smallest squared chordal distance is at least 2.704184. This cell of the table of best Grassmannian packings has no published value. The best value implied by known packings is 2.694758, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The value is checked with exact rational arithmetic from integer coordinates. We do not claim it is optimal.

1 Result

Theorem. *There are 17 subspaces of dimension 4 in $\mathbb{R}^{11}$ whose pairwise squared chordal distances $D(a, b) = 4 - \text{tr}(P_a P_b)$ are all at least*

$$D_{\min} = 2.704184 \dots$$

The exact value of $D_{\min}$, a rational number, is given in Appendix A.

2 Context

N. J. A. Sloane's *Table of Best Grassmannian Packings* (last modified 1997), now maintained by Henry Cohn, lists values for some N in the row $m = 11$, $n = 4$. The cell $N = 17$ is blank in Sloane's table, in Sloane's coordinate files and in Cohn's table as checked on October 8, 2026. The best value implied by known packings is 2.694758, obtained by deleting 1 of the 18 four-spaces of the packing found by this search. The Rankin simplex bound for this cell is $4(11 - 4)/11 \cdot 17/16 = 2.704545$, and the value above lies below it.

3 Verification

Each subspace is given by an integer $m \times n$ basis Y (coordinates times 10^{12} , rounded). With $G = Y^\top Y$ the projection is $P = YG^{-1}Y^\top$, so for two subspaces

$$\text{tr}(P_a P_b) = \frac{\text{tr}(\text{adj}(G_a) A \text{adj}(G_b) A^\top)}{\det G_a \det G_b}, \quad A = Y_a^\top Y_b,$$

an exact rational number. No orthonormality is assumed and no floating point is used. Run

`python grassmann/verify.py grassmann/packings/m11n4N17.txt`

in <https://github.com/Ge-limin/math719>.

*Found and checked by Claude, a model released by Anthropic, running on the author's laptop. The author sent the prompts.

4 How it was found

Many random starts of gradient descent on the Grassmannian against a log-sum-exp of the pairwise overlaps, sharpened over time, then a polishing stage (`grassmann/solve.py`); about 65 seconds on one laptop, with about ten such searches running at once.

A Exact value

numerator:

```
135209201822392656455927946693763981009347192017547412632221743278809978
173114047248274832382079339444522046785298970047747851847084574747926884
2305171870022529794247250308360131850233582129422
```

denominator:

```
4999999999999985178989161937481561542860815430648458500436530718346732464
437231211846821204559269172801898143782252262229281049434205789653607820
299828045100886361842743970330595098636494098143
```

B Coordinates

17 blocks of 11 rows with 4 integers each; divide by 10^{12} . File: `grassmann/packings/m11n4N17.txt`.

```
86838607432 -13345825223 394381911877 471167266847 -12253974268 173593714556 -133116126359 3560155768567963300588 -512003877087 -369851864313 115931499440
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